

Syed Shayan Mazahir

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EDUCATION

University of Waterloo

Expected June 2030

TECHNICAL SKILLS

Languages: Python, JavaScript, Java, Racket, C++ (exposure)

Frameworks & Libraries: Node.js, Express, Flask, Django, Discord.js, Discord.py

Research Interests: Natural Language Processing, Machine Learning, Human-Computer Interaction, AI-Assisted Code Generation

Tools & Technologies: Git/GitHub, MongoDB, PostgreSQL, REST APIs, OAuth, HTTPS/SSL, Linux/Unix

RESEARCH & PROJECTS

games.random - AI-Powered Code Generation Research

2024 – Present

- Developed full-stack application exploring automated code generation from natural language using large language models (Claude API), investigating challenges in structured generation including consistency, error handling, and semantic correctness
- Designed and evaluated prompt engineering techniques to ensure syntactically correct, executable JavaScript code with high reliability
- Implemented real-time validation pipeline and interactive Monaco-based code editor enabling human-in-the-loop refinement of AI outputs
- Open-sourced project to enable further research in AI-assisted game development and creative coding tools

Discord Bot Ecosystem - Scalable Community Management Systems

2018 – 2024

- Designed and deployed distributed bot architecture serving 5,000+ users across 15+ communities with 99.9% uptime
- Developed automated moderation algorithms using rule-based systems for content filtering and user behavior analysis
- Implemented persistent data storage systems (MongoDB, Quick.DB) optimized for high-frequency operations; contributed to Discord.js open-source community through documentation and code examples

EXPERIENCE

Louise Arbor Secondary School

2024 – 2025

Programming Hub Mentor & Research Assistant

- Led development of Flask-based web application with Google API integration serving 500+ students; conducted user experience research resulting in 60% improvement in student engagement
- Designed and delivered Python curriculum covering algorithms and data structures; mentored 20+ students, tracking learning outcomes and iterating on teaching methodologies
- Applied security best practices including SSL/HTTPS implementation and authentication protocols

Robotics Mentor & Algorithm Developer

- Developed and optimized pathfinding algorithms for autonomous robot navigation; led team of 15 students to First Place in RobOlympics
- Researched and implemented sensor fusion techniques for object detection using VEX Robotics platform

Aspect Systems

2020 – 2025

Community Operations & Infrastructure Lead

- Supported development of C++-based anti-cheat service analyzing player behavior patterns to detect exploits and security threats
- Designed automated workflow systems using Python for community management and incident response at scale
- Established documentation standards and knowledge base systems to scale operations across distributed team